

Simulation and Analysis Of Rectangular Waveguides

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Abstract

This report investigates the modal behavior and electromagnetic propagation characteristics of rectangular waveguides through a combination of analytical formulation and 3D full-wave simulation. Theoretical values for cutoff frequencies, propagation constants (β), and wave impedances were calculated using Microsoft Excel and validated against simulation data generated in Ansys HFSS. Spatial field distributions, including E-fields, H-fields, and Poynting vectors, were successfully visualized for the dominant TE₁₀ mode alongside higher-order modes (TE₂₀, TE₀₁, and TM₁₁) to confirm the theoretical power flow. Furthermore, waveguide operation below the cutoff frequency was modeled, mathematically and visually demonstrating the evanescent regime where energy becomes purely reactive and undergoes attenuation. Applying these principles, a custom waveguide was designed and simulated (20 x 10 mm) to guarantee strict single-mode operation across an 8 to 12 GHz target band. The results confirm a precise alignment between Maxwellian analytical theory and HFSS simulations, demonstrating the geometrical trade-offs required in engineering to isolate specific frequency bands, suppress destructive higher-order modes, and manage electromagnetic power propagation.

Introduction

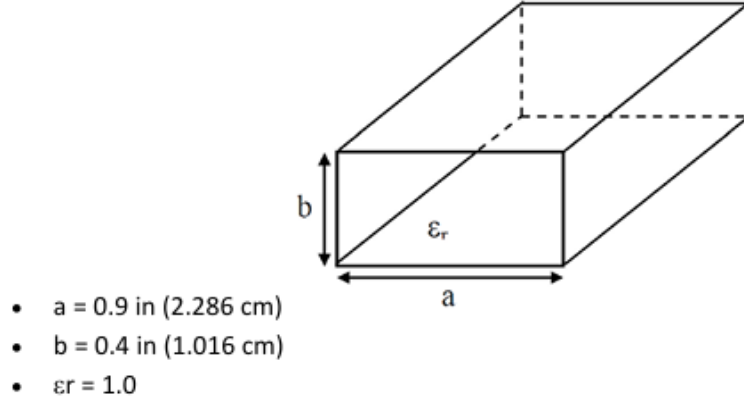
Rectangular waveguides are fundamental components in the field of electromagnetics, consisting of hollow metallic pipes designed to constrain and guide high-frequency EM waves. In modern RF and microwave systems such as radar, satellite communications, and high-power transmitters, waveguides are heavily utilized because they offer significantly lower attenuation and much higher power-handling capabilities compared to standard coaxial cables.

Because electromagnetic waves must satisfy strict boundary conditions against the walls of the waveguide, energy can only propagate in two distinct patterns: Transverse Electric (TE) modes, where the electric field is entirely transverse to the direction of propagation, and Transverse Magnetic (TM) modes, where the magnetic field is entirely transverse to the direction of propagation. Each mode is controlled by a cutoff frequency dictated by the physical dimensions of the waveguide, meaning the geometry acts as a natural high-pass filter.

This project integrates analytical electromagnetics with full-wave simulation (HFSS) to study rectangular waveguides. We will investigate modal behavior, field distributions, loss analysis, and design implications.

Waveguide Parameters

These parameters define the baseline geometry of the rectangular waveguide, this includes the dimensions of the planar opening, and the relative permittivity of the dielectric medium inside. According to the diagram below, the relative permittivity of the waveguide is 1, this means that the waveguide is air-filled.



Methodology

Analytical Formulation

To calculate the theoretical values for the cutoff frequencies, propagating constants (β), and wave impedances (Z_{TE} and Z_{TM}) across the swept frequency range, Microsoft Excel was utilized. The spreadsheet is programmed using the analytical formulas defined below to generate the datasets compared against the HFSS simulation results in Appendix A.

A wave of a given mode can propagate in a waveguide only if its frequency is greater than the cutoff frequency. The theoretical cutoff frequency for any TE or TM mode in a rectangular waveguide is:

$$f_{c,mn} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

The propagation constant (β) dictates the phase shift of the wave as it travels through the waveguide. The propagation constant must be real for propagation to occur. The propagation constant was found for each frequency step in each mode with the use of this equation:

$$\beta = \frac{2\pi f}{c} \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

The wave impedances for each mode type (Z_{TE} and Z_{TM}) link the E and H fields together. The wave impedance in Part 1 and Part 3 is found for each frequency step in each mode with the use of these equations:

$$Z_{TE} = \frac{\omega\mu}{\beta} = \frac{\eta}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

$$Z_{TM} = \frac{\beta}{\omega\epsilon} = \eta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

The free-space wavenumber (k) represents the phase shift per unit length of a wave traveling in an unbounded medium (like air). It is dependent on the operating frequency (ω) and the material properties (μ , ϵ):

$$k = \omega\sqrt{\mu\epsilon}$$

The cutoff wavenumber k_c is defined as:

$$k_c^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

Inside the geometry of the waveguide, the wave cannot propagate exactly as it does in free space. The actual phase shift per unit length down the guide is the propagation constant (β). These two values are linked by the cutoff wavenumber (k_c) through dispersion:

$$\beta = \sqrt{k^2 - k_c^2}$$

When the wavenumber is below the cutoff ($k < k_c$), the term under the square root becomes negative, making β imaginary. This proves that no real power propagates; the wave simply attenuates (evanescent decay).

At the cutoff ($k = k_c$), the propagation constant is exactly zero. The wave bounces perfectly up and down between the walls with no forward movement down the guide.

For TE modes, there is no electric field in the direction of propagation ($E_z = 0$). All other field components are derived from the magnetic field (H_z).

$$\begin{aligned} H_z &= H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \\ E_x &= \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \\ E_y &= -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \\ H_x &= -\frac{E_y}{Z_{TE}} \\ H_y &= \frac{E_x}{Z_{TE}} \end{aligned}$$

For TM modes, there is no magnetic field in the direction of propagation ($H_z = 0$). All other field components are derived from the electric field (E_z).

$$E_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$$

$$\begin{aligned}
E_z &= \frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \\
E_y &= -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z} \\
H_x &= -\frac{E_y}{Z_{TM}} \\
H_y &= \frac{E_x}{Z_{TM}}
\end{aligned}$$

HFSS Simulation Setup

Before drawing the model, the solution type must be defined. A Driven Modal solution type was chosen as it is used for calculating the mode-based S-parameters of passive, high-frequency structures such as a waveguide which are driven by a source.

The drawing was then modeled by placing a box with dimensions a and b along the Y and Z axis respectively. Since the waveguide is propagating in the X direction, the dimension of the waveguide along the X-axis is arbitrary, it was set to about 2 cm to represent a few wavelengths.

It was established that the waveguide is air filled (in a vacuum). So, the material was defined as “vacuum” from the material library. The “vacuum” definition in the library has a set relative permittivity of 1, an exact match to the diagram.

The boundary conditions of the waveguide must be assigned next, as they specify the field behavior at the edges of the problem region and object interfaces. Each face parallel to the direction of propagation (x-axis) was assigned a finite conductivity boundary to represent the conductivity losses.

After boundary condition assignments, the ports of the waveguide must be applied. The two faces perpendicular to the direction of propagation are assigned to a wave port. The wave port represents the surface through which the signal enters and exits the geometry. These ports are set up to have four modes each, where each mode has a matching integration line on both sides.

Integration lines define the voltage path used to calculate the characteristic impedance and establish the phase reference (0 degrees) for the S-parameters. These must be drawn along the path of the maximum electric field for each specific mode. For mode 1 (TE₁₀), the E-field is vertical and strongest in the exact center of the waveguide. For mode 2 (TE₂₀), the E-field is vertical but peaks at the $\frac{1}{4}$ and $\frac{3}{4}$ width marks. For mode 3 (TE₀₁), the E-field is horizontal and peaks in the center of the height. For mode 4 (TE/TM₁₁), the E-field is complex and radiates outward from the center.

Once all geometries of the waveguide were established, the mesh setting and convergence criteria were set. For a Driven solution type, the minimum value for the adaptive mesh frequency is $2/3^{\text{rd}}$ of the final frequency required. Since the analysis is done on the first four modes of the WR-90 waveguide, the cutoff frequency of the last mode is 16 GHz. So, adaptive mesh frequency should be slightly higher, at 20 GHz. The maximum number of passes was set to 10. This is the maximum number of mesh refinement cycles that HFSS will perform. The adaptive analysis will continue unless the convergence criteria are reached. The delta S parameter was changed to 0.002, which is the change in magnitude of the S-parameters between two consecutive passes.

Finally, a frequency sweep from 4 – 20 GHz with a step size of 0.5 GHz was added to the project. This sweep was set to discrete, with the start and stop frequencies set to 1 GHz and 20 GHz respectively. This simulation setup will be used to gather solution data and perform modal analysis.

Part 1: Modal Analysis

Mode Identification

An analysis of the first four propagating modes (sweep 4-20GHz) will be performed. To do this, the mode types (TE/TM) and cutoff frequencies shall be calculated using Excel.

To calculate the first four propagating modes, the defined dimensions of the waveguide (a and b), the dielectric material constant (in this case it is the speed of light), and the intrinsic impedance of free space (377 ohms) must be used.

Once these parameters are defined, a small table was created to identify the first four modes. For a standard waveguide where $a > b$, the first four modes are: TE₁₀ ($m = 1, n = 0$), TE₂₀ ($m = 2, n = 0$), TE₀₁ ($m = 0, n = 1$), TE₁₁ & TM₁₁ ($m = 1, n = 1$). Note that TE and TM modes share the same cutoff frequency for $m = 1, n = 1$.

The cutoff equation is then used in each of the first four modes, substituting m and n in each scenario. The analytical cutoff frequencies for the first four propagating modes are as follows:

TE ₁₀	6.56E+09
TE ₂₀	1.31E+10
TE ₀₁	1.48E+10
TE ₁₁ /TM ₁₁	1.62E+10

Results

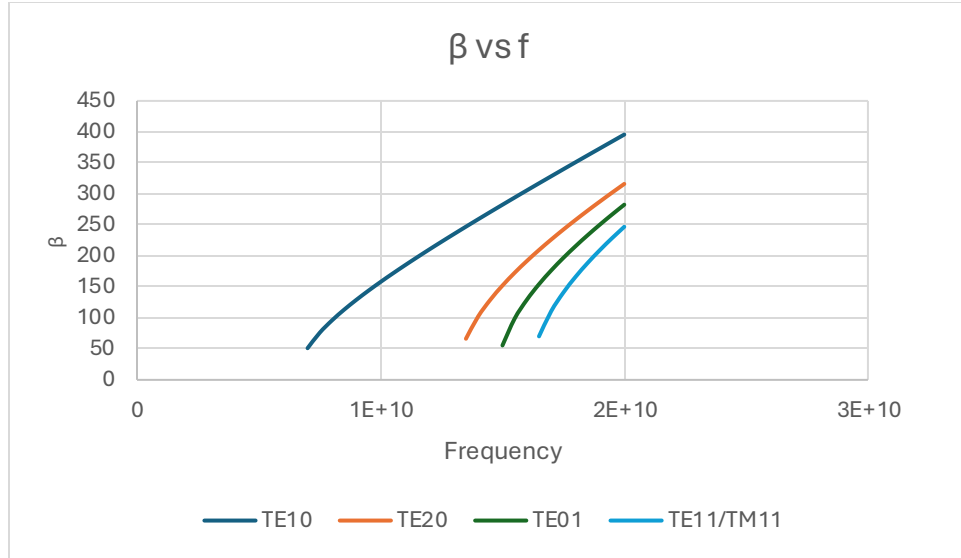


Figure 1

Figure 1 displays the propagation constant (β) plotted against frequency (4-20 GHz) for the first four modes, highlighting the transition at each respective cutoff frequency. Figure 2 plots the theoretical wave impedances (ZTE and ZTM) across the same sweep.

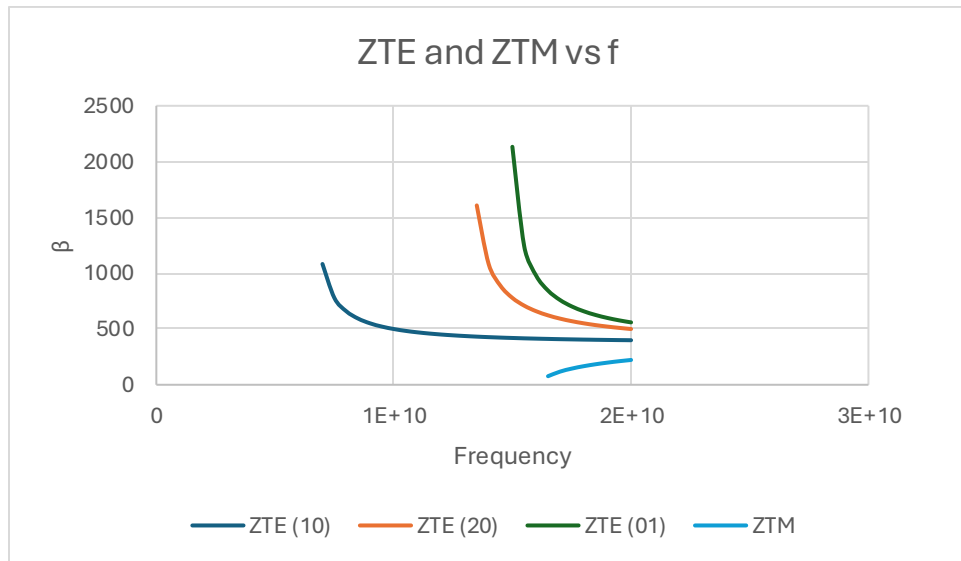


Figure 2

Figure 3 and Figure 4 show the HFSS full-wave simulation. These visualize the spatial field patterns and power concentrations within the waveguide cross-section.

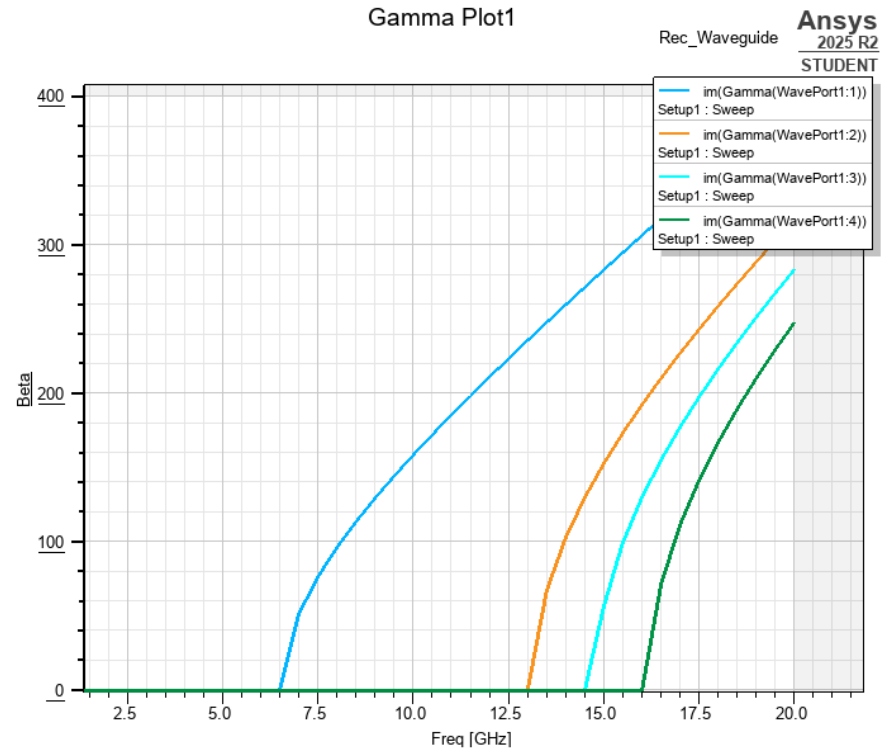


Figure 3

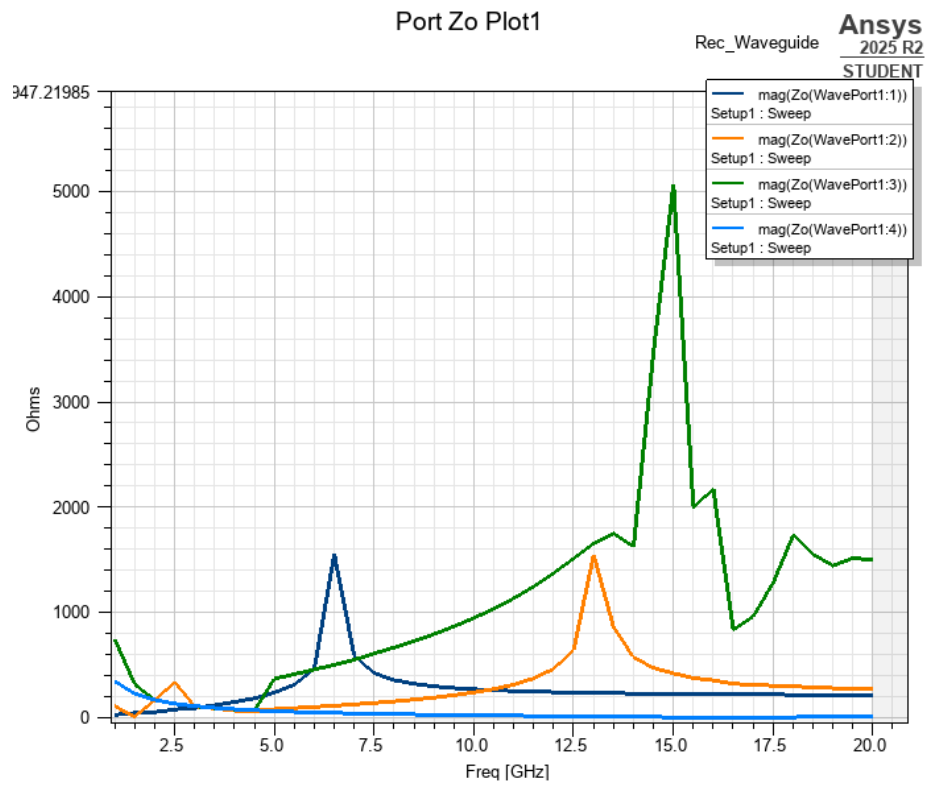


Figure 4

Discussion

A waveguide's dominant mode is characterized by the mode which has the lowest cutoff frequency. Based on the data, the TE₁₀ mode allows frequencies over 7 GHz to pass through, this is the lowest frequency to output a β value of all four modes. Another way to confirm the dominant mode comes from the characteristic dimensions of the waveguide. In a TE mode waveguide, if $a > b$, then the dominant mode is TE₁₀, otherwise if $a < b$, then the dominant mode is TE₀₁. Since $a > b$ in our waveguide, this confirms the dominant mode to be TE₁₀.

β is zero at the cutoff frequency and non-zero for frequencies above the cutoff. The HFSS simulation shows β increasing non-linearly with frequency, eventually approaching the free-space wave number (k). The simulated cutoff frequencies obtained from the HFSS model (Figure 3) shows a minor deviation of 0.5 GHz from the idealized analytical calculations (Figure 1). This discrepancy is attributed to the simulation's inclusion of finite wall conductivity and mesh-based calculations. Whereas the analytical model assumes an ideal, loss-less PEC structure. The alignment of these results confirms the analytical model.

When comparing the analytical results from Excell with the simulated HFSS data, there is a clear correlation, at the cutoff frequencies. Mathematically, the waveguide impedance for TE modes approaches infinity at the cutoff frequency. While the Excel model uses an idealized formula, HFSS calculates the impedance as the ratio of the transverse electrical and magnetic fields at the port interface. As the frequency approaches cutoff, the transverse magnetic field (H_t) approaches zero, which is the denominator in the impedance calculation. This leads to the sharp spikes observed in the simulation. These are not numerical errors; they are a direct result of the physics at the port boundary during the mode's transition from a fading state to a propagating state.

Below the cutoff for each mode, the simulation produces some interesting behavior. This is expected, as the mode is non-propagating in this region, and the solver struggles to resolve a stable field ratio. However, as the frequency increases, the wave impedances stabilize toward the free-space intrinsic impedance of 377Ω . This confirms that at higher frequencies, the waveguide dimensions have less influence on impedance, and both models accurately reflect the propagation behavior.

A final comparison was made between the free-space wavenumber (k) and the simulated propagation constant (β). Theoretically, k represents the phase shift per unit length in an unbounded medium (air), while β accounts for the phase shift within the confined geometry of the waveguide.

The simulation results confirm that β is always less than k at any given frequency above cutoff. This indicates that the guide wavelength is longer than the free-space wavelength. At the cutoff frequency, β is zero, meaning no phase progression occurs along the guide. As the frequency increases toward 20 GHz, the curve for β begins to approach the linear slope of k . This shows that as the frequency gets much higher than the cutoff, wave 'feels' the walls of the waveguide less and begins to behave more like a wave propagating in free space.

Part 2: Field Distribution and Power Flow

Field Plots

E-field Distribution:

Viewed from the top-down (X-Y plane), Figure 5 shows the E-field for Mode 1. It shows a series of high-intensity (red) bands separated by zero-intensity (blue gaps). The bands stretch uniformly across the width of the waveguide but fade to zero at the side walls. This pattern repeats down the length (X-axis) of the waveguide.

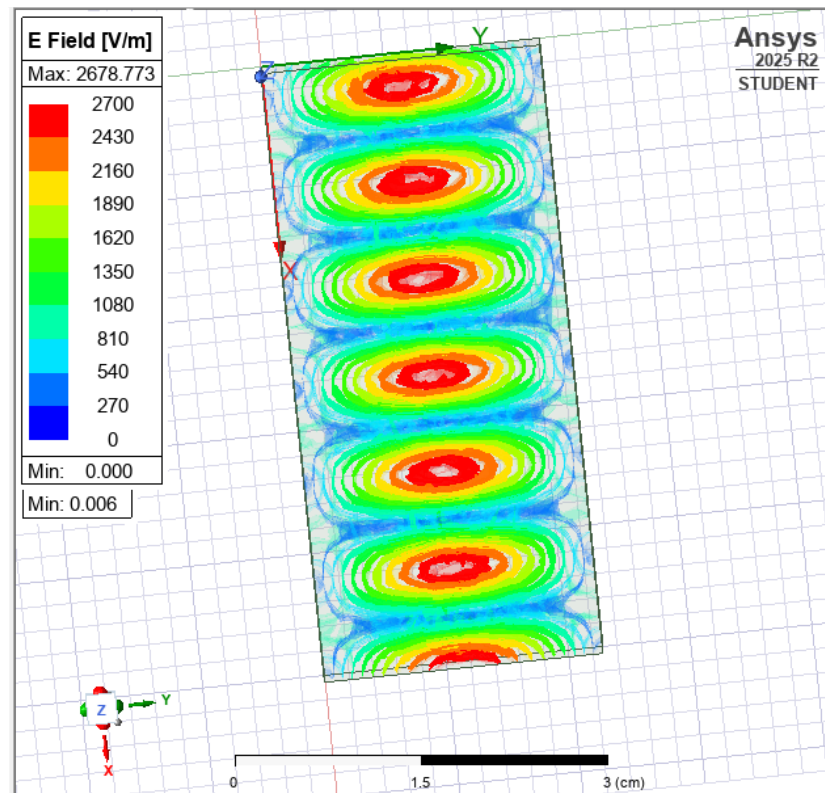


Figure 5 – TE₁₀

Figure 6 shows the E-field for Mode 2. The field is split into two parallel lanes of high-intensity regions traveling down the length of the waveguide. A continuous blue “null” line runs exactly down the center axis from the port, separating the two lanes.

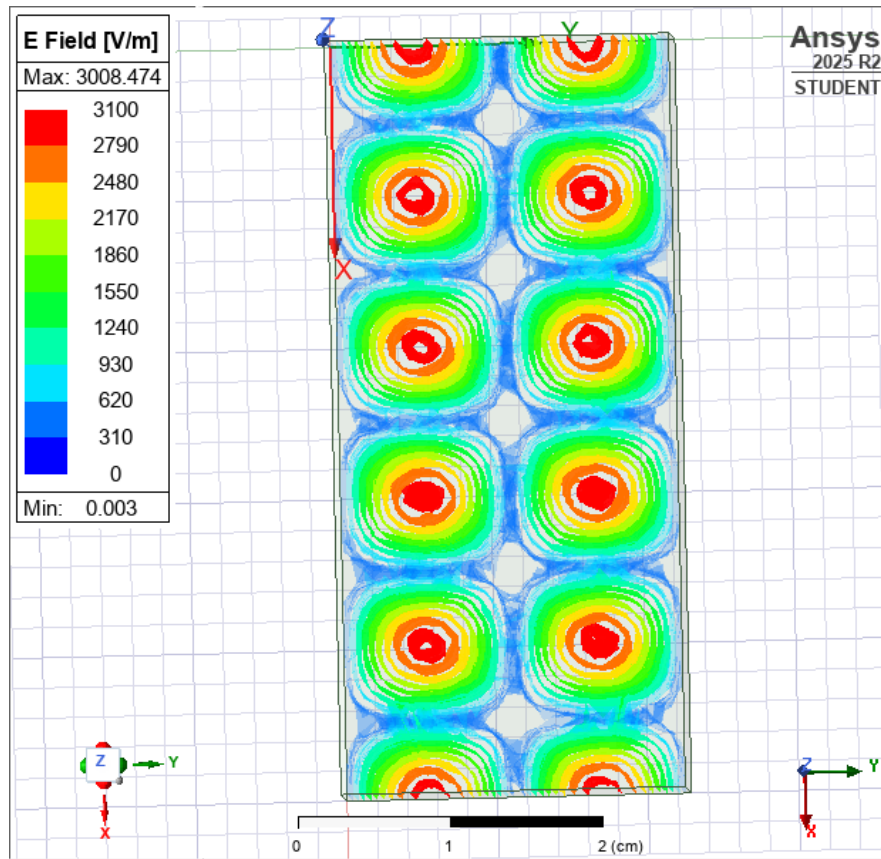


Figure 6 – TE₂₀

Viewed from the side profile (X-Z plane), Figure 7 shows the E-field for Mode 3. This shows a 90-degree shift in behavior. The E-field’s high-intensity areas now span horizontally (along the Y-axis). These areas are centered between the top and bottom walls and repeat periodically down the

length of the waveguide. The field intensity drops completely to zero (blue) at the top and bottom boundaries.

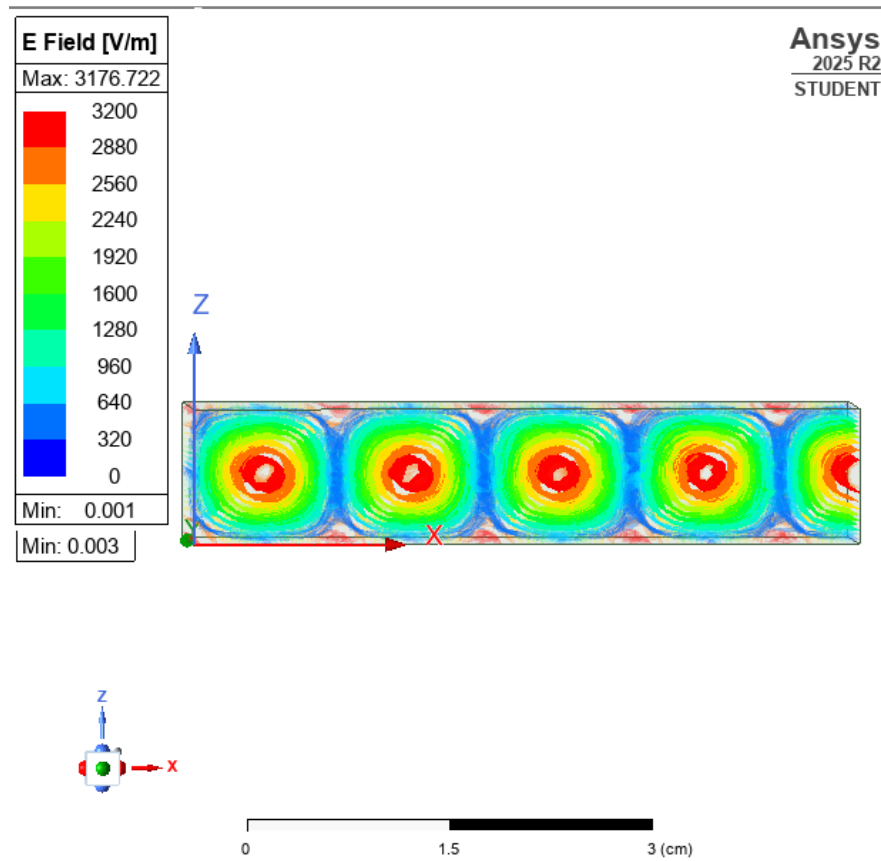


Figure 7 – TE01

Figure 8 shows the E-field for Mode 4. It's a highly complex, standing-wave pattern. Instead of simple parallel bands, the high-intensity regions form pockets that vary in both width and height (Y and Z-axis respectively) as they repeat down the length of the guide.

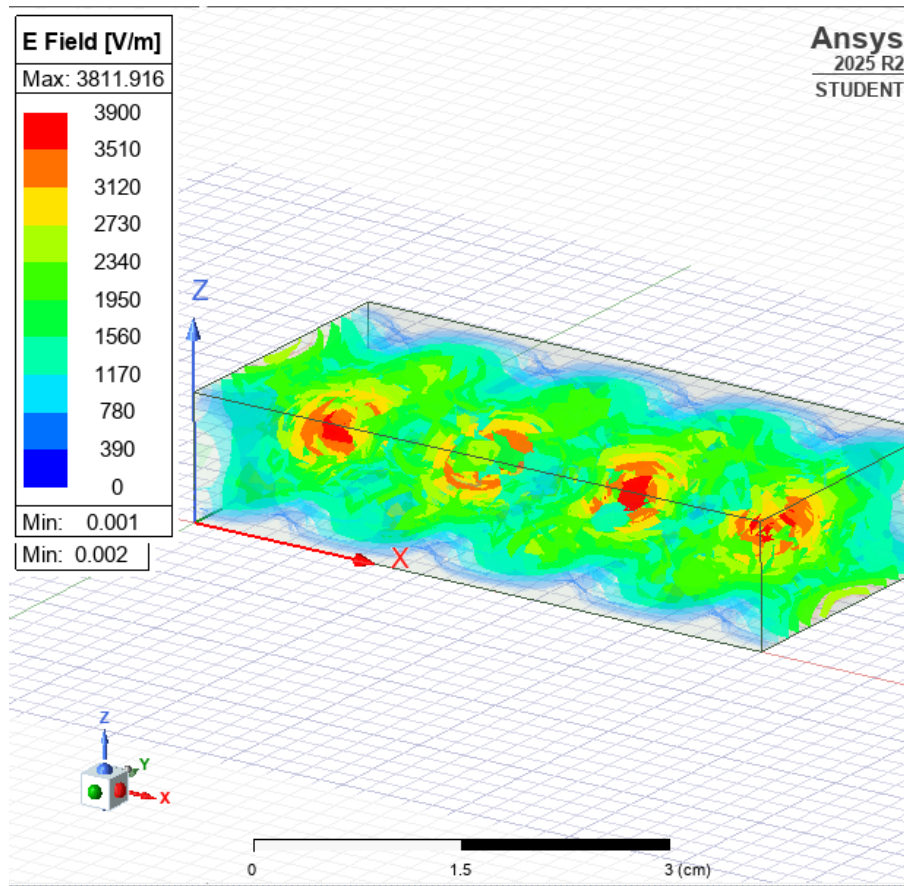


Figure 8 – TM11

H-Field Distribution:

Viewed from the top-down (X-Y plane), the H-field in Figure 9 forms a wide, oval-shaped band of high-intensity (red) that stretches across the center of the waveguide. These bands repeat periodically down the length of waveguide (X-axis). The field intensity drops to blue on the left and right walls.

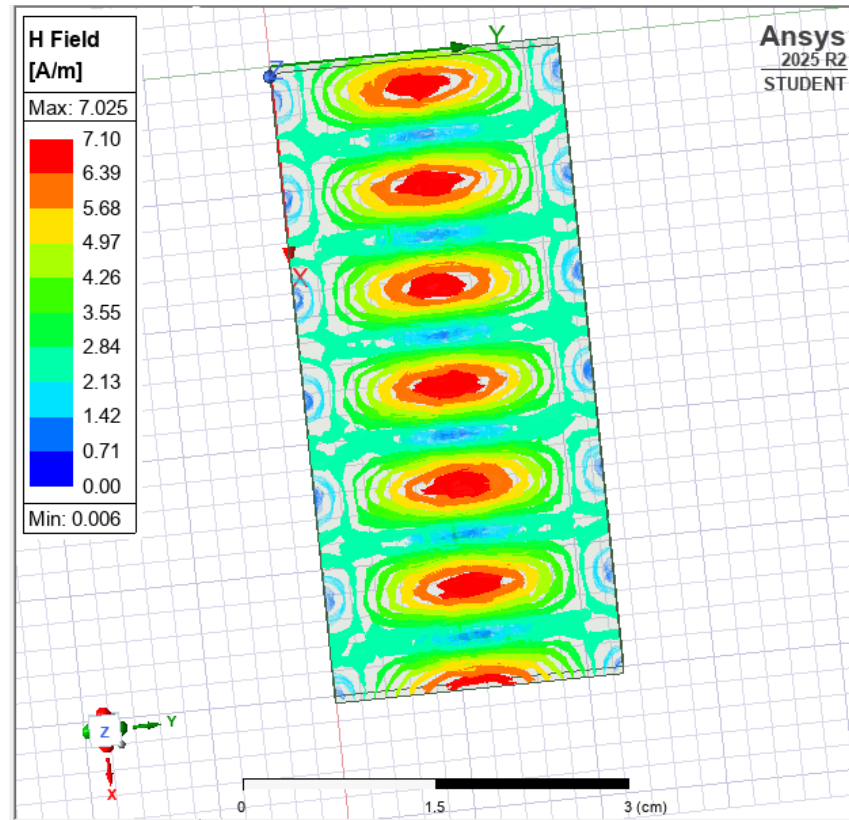


Figure 9 – TE10

Viewed from the top-down, the H-field in Figure 10 forms a repeating “checkerboard” pattern. The field is broken into distinct high-intensity regions located directly on the top and bottom boundaries, as well as a row directly down the center axis. Between these regions are two more

rows of high-intensity nodes (at the quarter and three-quarter width marks). These rows are shifted longitudinally along the X-axis, alternating perfectly with the boundary and center nodes.

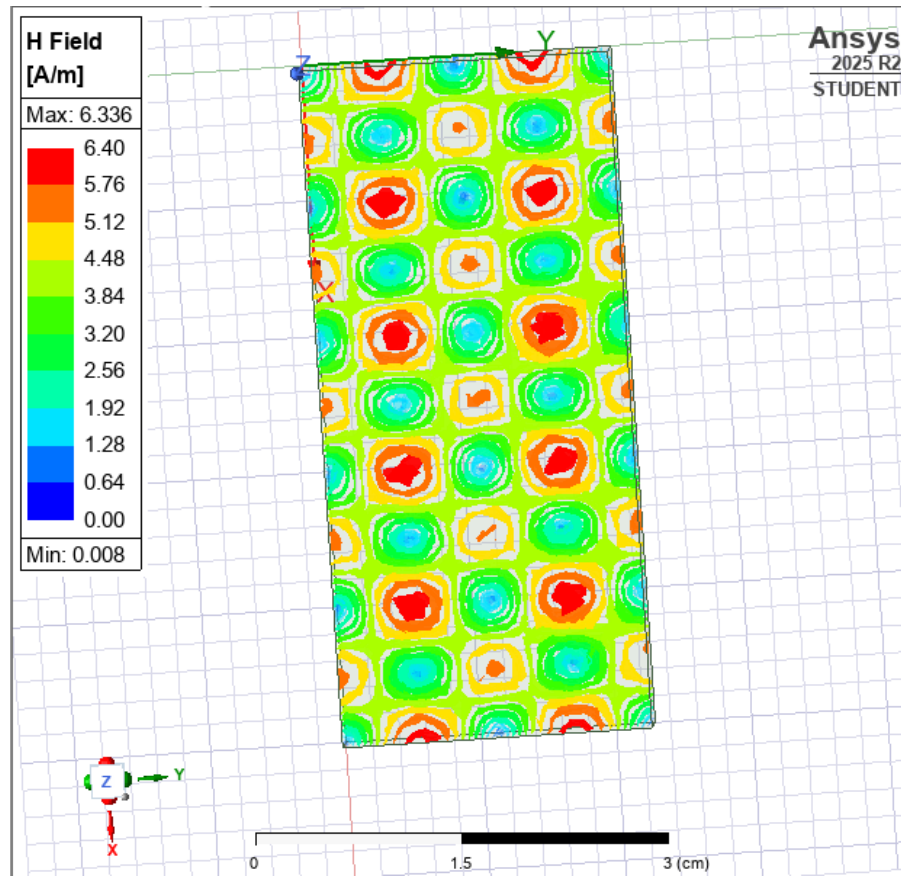


Figure 10 – TE₂₀

Viewed from the side profile (X-Z plane), the H-field in Figure 11 forms an alternating, staggered pattern. The highest intensity regions are pinned directly against the top and bottom boundaries. Traveling down the center height of the waveguide is a secondary row of high-intensity (orange) nodes. These central nodes are offset along the X-axis from the boundary nodes, with low-intensity gaps in between.

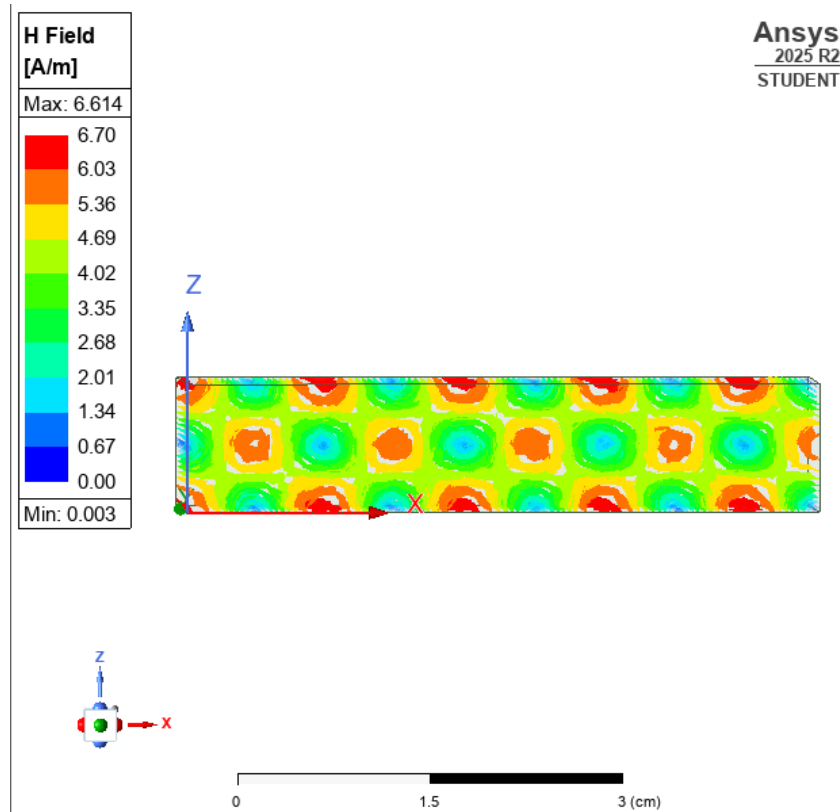


Figure 11 – TE₀₁

Much like its E-field counterpart, this Mode 4 pattern is very different for the other TE modes. The H-field forms a series of connected ring-like structures that pulse periodically down the length of the waveguide. The center of these rings maintains a low intensity (blue) "hole" running down the central axis. The highest intensity regions form the outer edges of these rings, which press against the top, bottom, and sides of the boundaries.

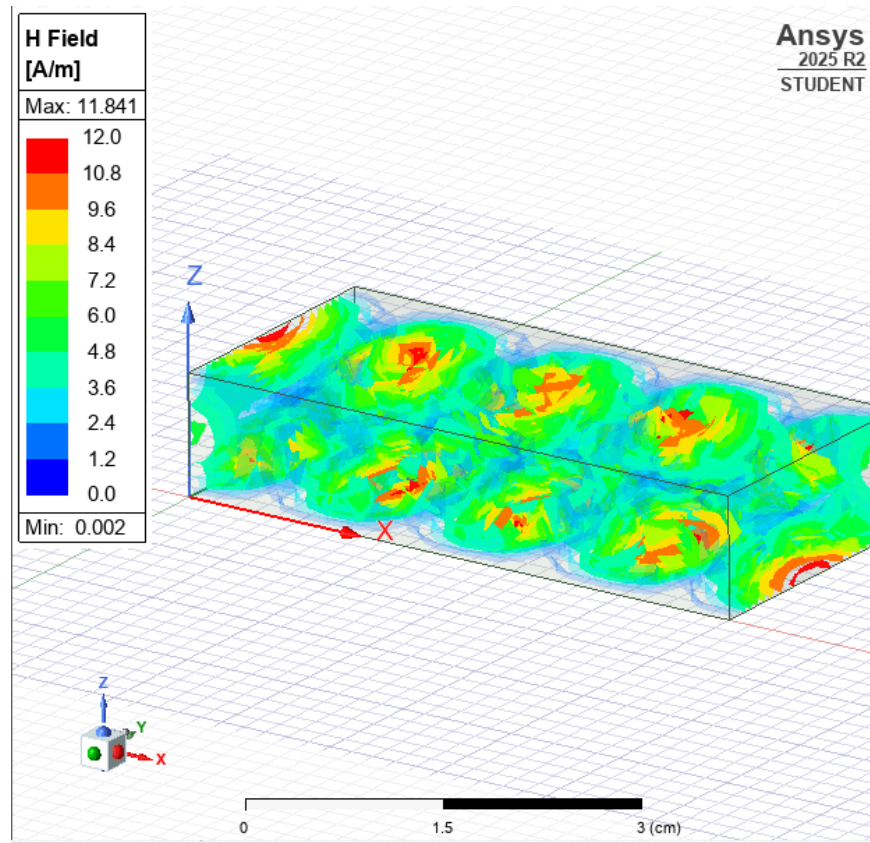


Figure 12 – TM₁₁

Poynting Vector:

Figure 13 shows the Poynting vector plot for Mode 1. It shows a stream of vectors pointing down the length of the waveguide (along X-axis). The highest power density is concentrated in a single column running down the central axis. The power density drops to near zero at side boundaries.

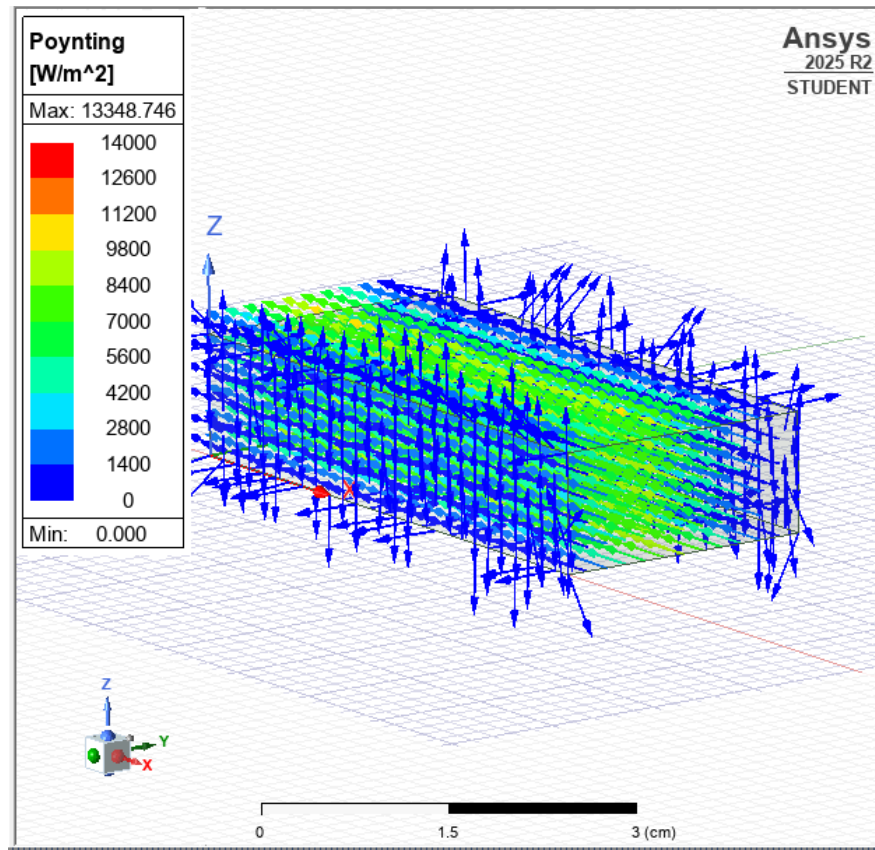


Figure 13 – TE10

The Poynting vector plot for Mode 2 shows vectors pointing down the length of the waveguide, split into two distinct parallel lanes. The vectors drop to zero intensity between the lanes and at the left and right sides of the boundaries.

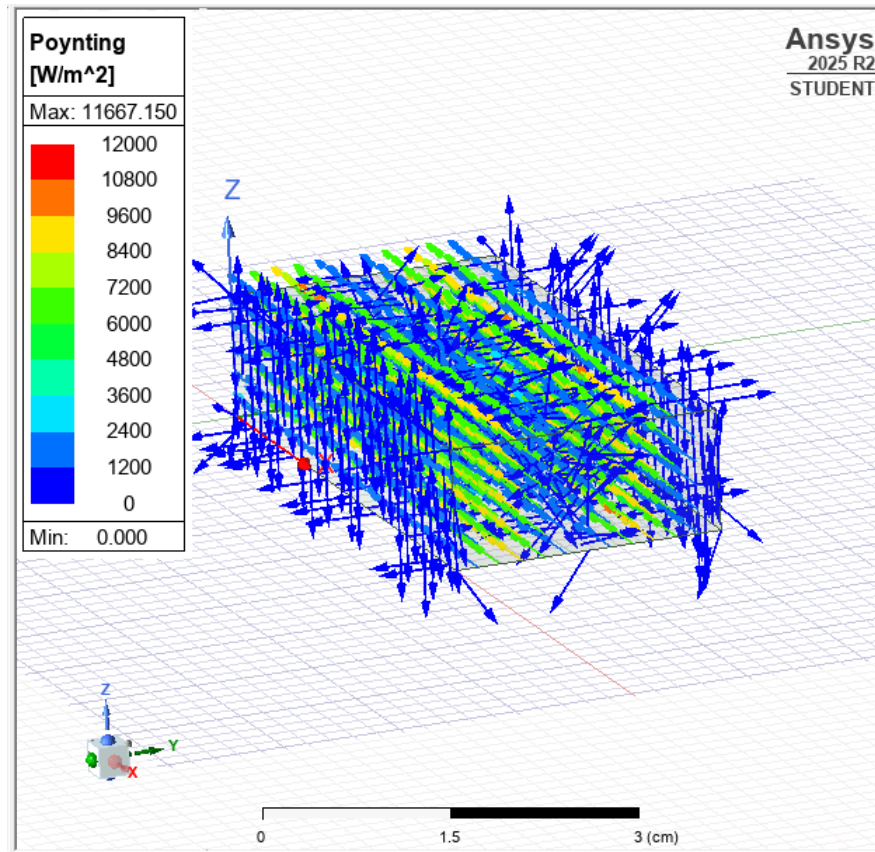


Figure 14 – TE₂₀

Figure 15 shows the Poynting vector plot for Mode 3, pointing down the waveguide (along X-axis). The highest power density forms a wide, flat horizontal sheet centered between the top and bottom walls. The vector magnitude drops to zero at the upper and lower boundaries.

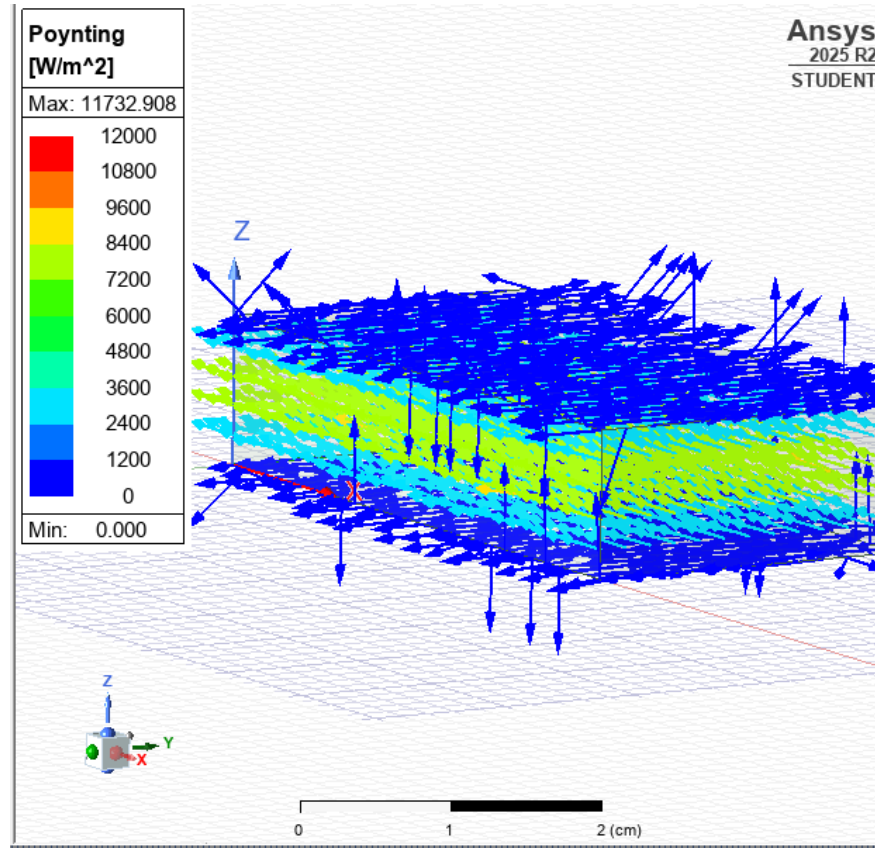


Figure 15 – TE₀₁

The Poynting vector plot for Mode 4 shown in Figure 16 has vectors pointing along the X-axis down the waveguide. However, the spatial distribution of power is completely hollow. The exact center axis is a dead zone indicating zero power flow. The highest power density is pushed outward, forming two lobes located on the top and bottom walls of the waveguide.

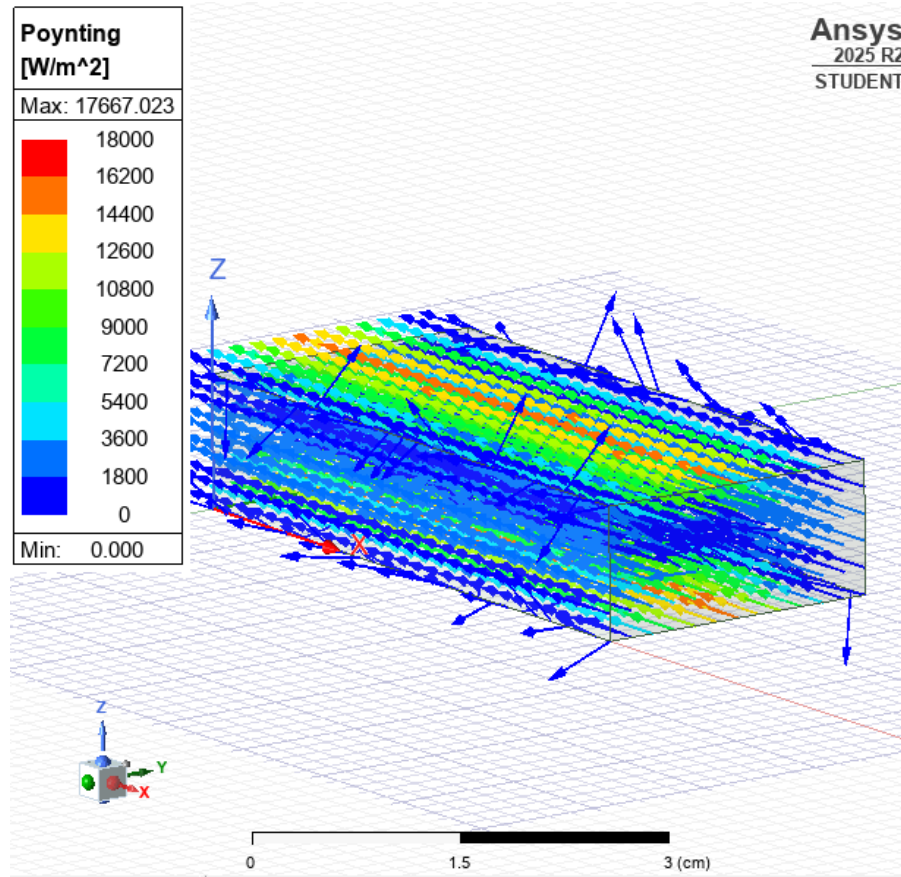


Figure 16 -TM11

Discussion

Mode 1: TE10

The E-field distribution in Figure 5 visualizes the TE10 mode actively propagating. The single red band across the width confirms the ‘1’ index $\sin(\pi x/a)$ function. This sine wave creates a single peak in the center and forces the field to zero at the boundaries ($x = 0$ and $x = a$). By looking down the the length of the guide, we can physically see the wave’s peaks and nulls. The distance between two consecutive red peaks represents one-half of the guide wavelength ($\lambda/2$). Because the field stays concentrated in the center, the power flows straight down the middle of the wavelength like a continuous pulsing channel.

The H-field distribution in Figure 9 validates the theoretical solution. The single band of intensity across the width confirms the analytical $m = 1$ index. Because the E-field was purely vertical, the H-field must be purely horizontal (transverse E-field) so that they are 90 degrees apart. The color magnitude follows the exact same spatial pattern as the E-field because the two are proportionally linked by the wave impedance. When you take the cross-product of a vertical E-field and horizontal H-field, the resulting vector (power) flows straight down the remaining axis of the waveguide.

The Poynting vector represents the directional power density of the EM wave. Figure 13 links the E and H field for TE₁₀ mode. In the dominant mode (TE₁₀), the E-field is vertical and peaks in the center, and the transverse H-field is horizontal and peaks in the center. Taking the cross-product of the two fields results in the Poynting vector that points in the direction of propagation (+X). Because the power density is proportional to product of the field magnitudes, this plot confirms that most real power transfer in the TE₁₀ mode travels straight down the center channel of the waveguide.

Mode 2: TE₂₀

The effect of increasing the mode number ($m = 2$) is visible in Figure 6. The wave propagates as two distinct, parallel channels of energy. The analytical expression for TE₂₀ includes a $\sin(2\pi x/a)$ term, which predicts this exact zero-crossing (null) at the center ($x = a/2$). The zero-intensity null perfectly splits the waveguide in half horizontally. Because the two sides are 180 degrees out of phase, as a high-intensity peak forms on the left lane, the right lane will also have a peak but pointing in the opposite vertical directions. In terms of power distribution, increasing this mode number effectively splits the power flow into two distinct parallel channels, rather than the single central channel seen in TE₁₀.

The H-field in Figure 10 shows the $m = 2$ variation for the magnetic field. The alternating rows represent the two different directional components of the H-field. The transverse magnetic field (H_y) drives the intensity in the two quarter-width lanes. Conversely, the magnetic field (H_x) drives the intensity at the walls and the exact center of the waveguide. Because these two components are 90 degrees out of phase, their maximums do not overlap. Instead, they take turns reaching peak intensity as the wave propagates, creating the dense, 3D lattice of magnetic energy.

The Poynting vector shown in Figure 14 perfectly illustrates the power flow for the TE₂₀ mode. The vector is calculated via the cross product of the electric and magnetic fields. Because both the transverse E-field and transverse H-field drop to zero exactly at the center of the waveguide due to the $m = 2$ index, their resulting cross product must also be zero at the center. This plots shows the total electromagnetic power is divided into two equal, parallel streams traveling down the guide.

Mode 3: TE₀₁

This distribution in Figure 7 aligns with the analytical equations for the TE₀₁ mode. Here, the $m = 0$ index means there is no variation across the width, and the $n = 1$ index introduces a single $\sin(2\pi y/b)$ variation across the height. Because the E-field is strongest in the middle horizontal plane, the power distribution follows suit, spreading out into a wide horizontal sheet rather than a vertical column.

The distribution in Figure 11 emulates the nature of the magnetic field for the TE₀₁ mode. Unlike the electric field, which must drop to zero at the metal boundaries, the tangential magnetic field is

strongest at the conductive walls. The red nodes along the top and bottom represent the longitudinal magnetic component (H_x), which peaks at the boundaries. The orange nodes in the center represent the transverse magnetic component (H_z), which peaks in the middle of the guide. Because these two components are 90 degrees out of phase, their maximums do not overlap, creating the alternating “checkerboard” magnitude pattern observed as the wave propagates.

The Poynting vector plot in Figure 15 accurately visualizes the power transfer for the TE01 mode. Because the transverse E-field is strictly horizontal, and the transverse H-field is strictly vertical, their cross-product results in a Poynting vector straight down the waveguide. Since both fields reach their maximum intensity in the middle of the guide’s height (from the $n = 1$ index), the resulting power flow is concentrated in the same horizontal plane. Although the TE10 mode pushed power through a vertical column, this plot proves that the TE01 mode propagates energy as a flat horizontal plane. This behavior is representative of the 90-degree phase shift between the two modes.

Mode 4: TE11

The distribution in Figure 8 demonstrates the effects of increasing both mode numbers simultaneously ($m = 1, n = 1$). In a rectangular waveguide, the TE11 and TM11 modes share the exact same theoretical cutoff frequency ($f_c = 16.15$ GHz), making them “degenerative modes.” Because they calculate at the same frequency, HFSS solver might swap which is labeled “Mode 4.” Unlike the TE modes, the transverse E-field for a TM mode drops to zero in the center and radiates outward to terminate perpendicular on the walls. The power distribution here would form a ring shape, pushing away from the center.

The distribution in Figure 12 demonstrates the complex volumes when both indices are non-zero ($m = 1, n = 1$). By definition, a TM mode has no longitudinal magnetic field component ($H_x = 0$), meaning all magnetic field lines must form closed circulating loops in the transverse plane. As the wave propagates, these circulating fields pulse in intensity, creating 3D donut-shaped magnitude structures. The “hole” in the center of these donuts is the continuous central null where the transverse magnetic fields cancel out. This confirms that energy travels as a sequence of pulsing magnetic rings rather than a solid central core.

The visualization in Figure 16 ties together the complex E and H fields in TE11 mode. The Poynting vector dictates that power only flows where the electric and magnetic fields exist. Because both the transverse fields drop to zero in the center of the waveguide for this mode, the resulting cross product must also be zero in the center. Consequently, the electromagnetic power is forced away from the center axis, breaking up the energy flow into a hollow path. This is a huge contrast to the dominant TE10 mode, which concentrates all its power directly in the center, proving how higher-order modes drastically change the transmission of energy.

Part 3: Geometry Modification

Analytical Design

A new design of the waveguide was implemented such that the a dimension was modified for a TE₁₀ mode cutoff of 10 GHz. The cutoff frequency for the dominant TE₁₀ mode in a rectangular waveguide is controlled by the width of the broad wall (a). To redesign the waveguide to achieve a specific cutoff frequency, the theoretical formula can be arranged to solve for the required dimension a :

$$f_{c,10} = \frac{c}{2a}$$

$$a = \frac{c}{2f_{c,10}}$$

Assuming the waveguide is air-filled ($c = 3 \times 10^8$ m/s), the new dimension is calculated as:

$$a = \frac{3 * 10^8}{2(10 * 10^9)}$$

$a = 0.015$ meters

Therefore, the width of the waveguide must be modified to exactly 15 mm to shift the dominant mode cutoff to 10 GHz.

HFSS Validation

The new calculated dimensions were used to modify the waveguide in HFSS to confirm the new cutoff frequency for the TE₁₀ mode.

Figure 17 presents the HFSS simulated propagation constant (β) for the modified waveguide across a swept frequency range of 0 – 15 GHz. As predicted by the analytical design calculations, the dominant mode (blue trace, Mode 1) exhibits a constant of zero until the frequency reaches 10 GHz.

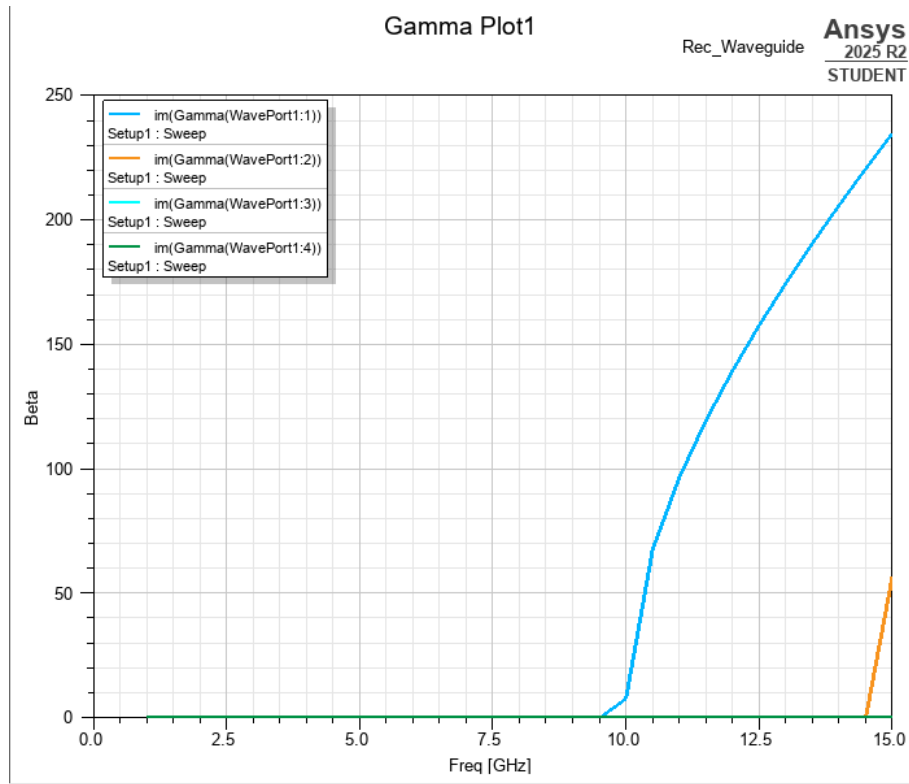


Figure 17

Discussion

While the math shows β becoming imaginary below the cutoff frequency, the physical meaning of a complex propagation constant is a termination of real power transfer.

When β is a real number above cutoff, it represents a spatial phase shift, where the wave physically propagates down the guide. However, when β becomes complex, the waveguide stops acting like a transmission line. Physically, a complex β means electromagnetic energy is reactive. The fields still exist in the input port, but because they are exponentially decaying instead of oscillating, no real energy travels down the guide. Instead, this energy is reflected toward the source. In this evanescent state, the waveguide physically behaves more like a giant inductor or capacitor rather than a transmission line.

Part 5: Open-Ended Design Challenge

Design Goal

The objective of the design challenge is to create a waveguide that guarantees a single-mode propagation across a frequency band of 8 – 12 GHz. This requires the dominant mode (TE₁₀)

cutoff frequency to be below 8 GHz, and the next higher-order mode cutoffs (TE20 and TE01) to be above 12 GHz.

Justification

Custom dimensions of $a = 20$ mm and $b = 10$ mm were selected to satisfy these requirements. The theoretical cutoff frequencies for these dimensions are calculated below:

Dominant Mode (TE10):

$$f_{c,10} = \frac{3 * 10^8}{2(0.02)} = 7.5 \text{ GHz}$$

Second Mode (TE20):

$$f_{c,20} = \frac{3 * 10^8}{2(0.01)} = 15 \text{ GHz}$$

Third Mode (TE01):

$$f_{c,01} = \frac{3 * 10^8}{2(0.01)} = 15 \text{ GHz}$$

The TE10 mode begins propagating at exactly 7.50 GHz, which satisfies the requirement to operate below 8 GHz. The next available modes do not begin propagating until 15 GHz. Therefore, the entire 8 – 12 GHz band is guaranteed to be strictly single-mode region.

HFSS Validation

Figure 18 confirms the analytical calculations. Between 8 GHz and 12 GHz, only Mode 1 (blue) exhibits a real, positive propagation constant (β). Modes 2 and 3 remain a $\beta = 0$ (evanescent) throughout this entire band.

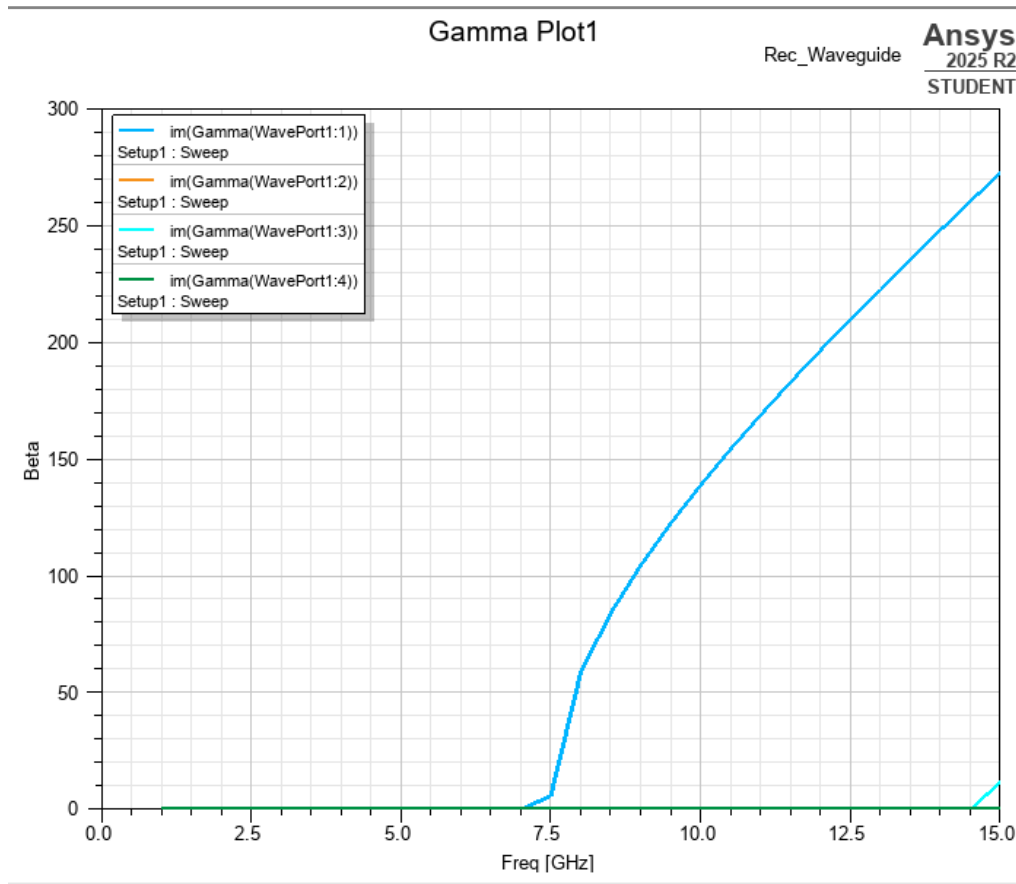


Figure 18

Discussion

Selecting the dimensions a and b requires balancing physical and electrical constraints to isolate the target band:

1. **Width (a) Trade-offs:** The dimension a must be exactly 20 mm to fix the TE₁₀ cutoff at 7.5 GHz to ensure propagation at the target frequency (8 GHz). However, this width directly dictates the TE₂₀ cutoff as well. Because the TE₂₀ mode is perfectly double the TE₁₀ frequency in a rectangular guide, setting $a = 20$ mm permanently locks the TE₂₀ mode cutoff at 15 GHz. This successfully keeps the higher-order mode out of the 8 – 12 GHz band without causing multi-mode interference.
2. **Height (b) Trade-offs:** The dimension b must be kept small enough to push the TE₀₁ cutoff above the 12 GHz target. However, shrinking the height reduces the total cross-section area of the waveguide. A smaller physical area reduces the peak power-handling capacity before breakdown occurs and slightly increases losses in the walls. The height of 10 mm excludes the TE₀₁ mode from operating in the target frequency band while keeping the most remaining area for power transmission.

Results Summary Table

Mode	Theoretical Cutoff (f_c)	Propagation Constant (β)	Wave Impedance (Z)	Observations
TE ₁₀	~ 6.56 GHz	$\beta = 0$ below cutoff frequency. Increases just above cutoff, toward free-space wavenumber k .	Starts near infinity at cutoff. Decreases toward free-space impedance (377 Ω)	E-field is vertical. Power flows in a single concentrated column straight down the center of the broad wall
TE ₂₀	~ 13.12 GHz	$\beta = 0$ below cutoff frequency. Same asymptotic trend as Mode 1 once propagating.	Starts near infinity at cutoff. Decreases toward free-space impedance (377 Ω)	Field splits into two parallel channels. Zero-intensity null directly down the center axis.
TE ₀₁	~14.80 GHz	$\beta = 0$ below cutoff frequency. Same asymptotic trend once propagating.	Starts near infinity at cutoff. Decreases toward free-space impedance (377 Ω)	90-degree spatial shift from TE ₁₀ . Power distributes into a wide, flat horizontal sheet.
TM ₁₁	~ 16.2 GHz	$\beta = 0$ below cutoff frequency. Same asymptotic trend once propagating.	Starts at 0 Ω at cutoff. Increases toward free-space impedance (377 Ω)	$H_x = 0$ forces circulating magnetic loops. Power avoids the center, flowing in 3D hollow donut rings.

Conclusion

This experiment provided a comprehensive analysis of electromagnetic wave propagation in rectangular waveguides. Throughout the analysis, the spatial characteristics of the dominant TE₁₀ mode and higher-order modes (TE₂₀, TM₁₁ and TE₀₁) were evaluated. Visualizing the E-fields, H-fields, and Poynting vectors confirmed that power distribution is heavily dependent on the mode indices. While the dominant mode concentrates energy in a single central channel, higher-order modes split into parallel, horizontal, or hollow channels. Furthermore, the look into evanescent behavior in Part 3 physically demonstrated that operating below the cutoff frequency results in reactive energy and exponential attenuation without real power transfer.

An important objective of this analysis was to validate analytical Maxwell equations against numerical 3D simulations. The data collected from HFSS – specifically the cutoff frequencies, propagation constants, and wave impedances – aligned perfectly with the theoretical models calculated via analytical formulas. The dispersion plots confirmed the asymptotic behavior of β approaching the free-space wavenumber (k) at high frequencies. The also confirmed wave impedances converging toward the intrinsic impedance of free space (η_0) proving the reliability of the simulation software.

The most significant takeaway from this analysis was understanding the practical engineering trade-offs required in waveguide design. Part 5 reinforced that a waveguide's dimensions cannot be chosen arbitrarily. Modifying the width (a) and height (b) to isolate the 8 –12 GHz band for single-mode operation required careful considerations: ensuring the TE₁₀ cutoff was low enough to allow propagation, while keeping the waveguide narrow and short enough to push the TE₂₀ and TE₀₁ cutoffs safely above the operation band. This experiment highlighted the constant compromise between physical design and electrical limits in engineering.

Appendix

A: Excel Formulas

The following table details the Microsoft Excel formulas used to calculate the theoretical waveguide parameters. The spreadsheet evaluated these formulas across a swept frequency column to generate the theoretical plots compared to the HFSS data.

Parameter	Excel Formula Example	Description
Constants Setup	\$B\$1 = 0.02286, \$B\$2 = 0.01016, \$B\$3 = 3E8	Storing dimensions a , b , c
Frequency (f)	A15 = 4.00E9 A16 = A15 + 0.50E9	Swept frequency input column
Cutoff Freq. (f_c)	=(B3/2$)*SQRT((m/B1$)^2 + (n/B2$)^2)	Calculate theoretical cutoff. Substitute m and n for specific mode
Propagation Const. (β)	=IF(A15>\$B\$7, (2*PI()*A15/\$B\$3)*SQRT(1-(\$B\$7/A15)^2), NA())	Set $\beta = 0$ below cutoff, calculate β for each frequency step
TE Wave Impedance (ZTE)	=IF(A15>\$B\$7, \$B\$4/SQRT(1-(\$B\$7/A15)^2), NA())	Calculates $\omega\mu / \beta$
TM Wave Impedance (ZTM)	=IF(A15>\$B\$10, \$B\$4*SQRT(1-(\$B\$10/A15)^2), NA())	Calculates $\beta / \omega\epsilon$

B: Additional Plots

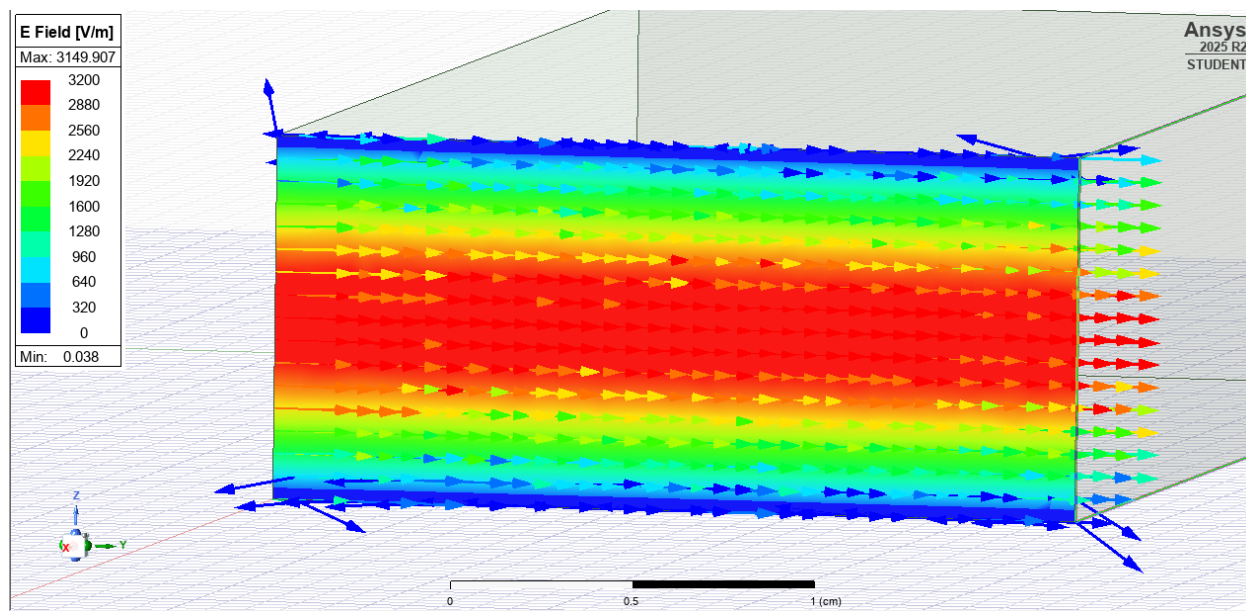


Figure 19 – Port Face E-field distribution Mode 1

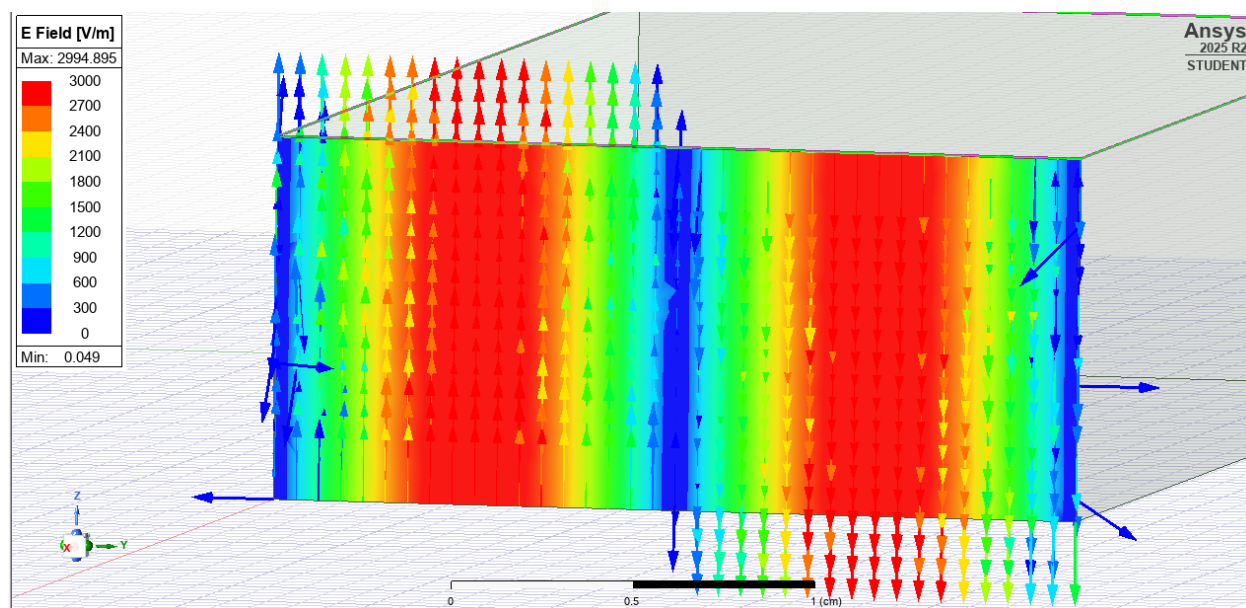


Figure 20 – Port Face E-field distribution Mode 2

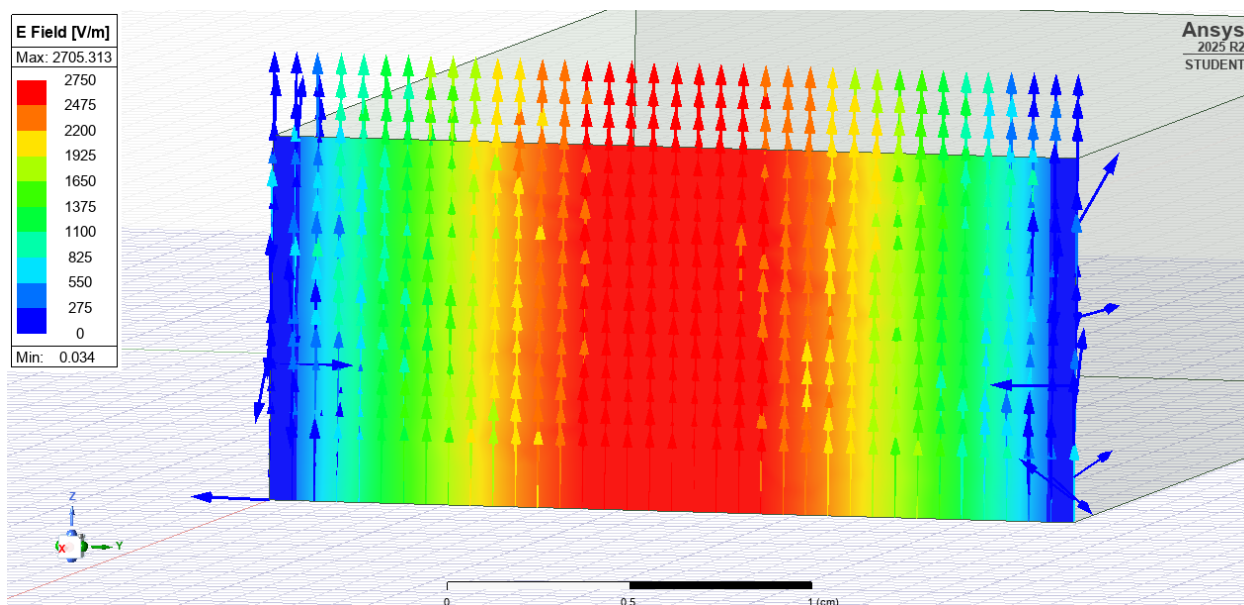


Figure 21 - Port Face E-field distribution Mode 3

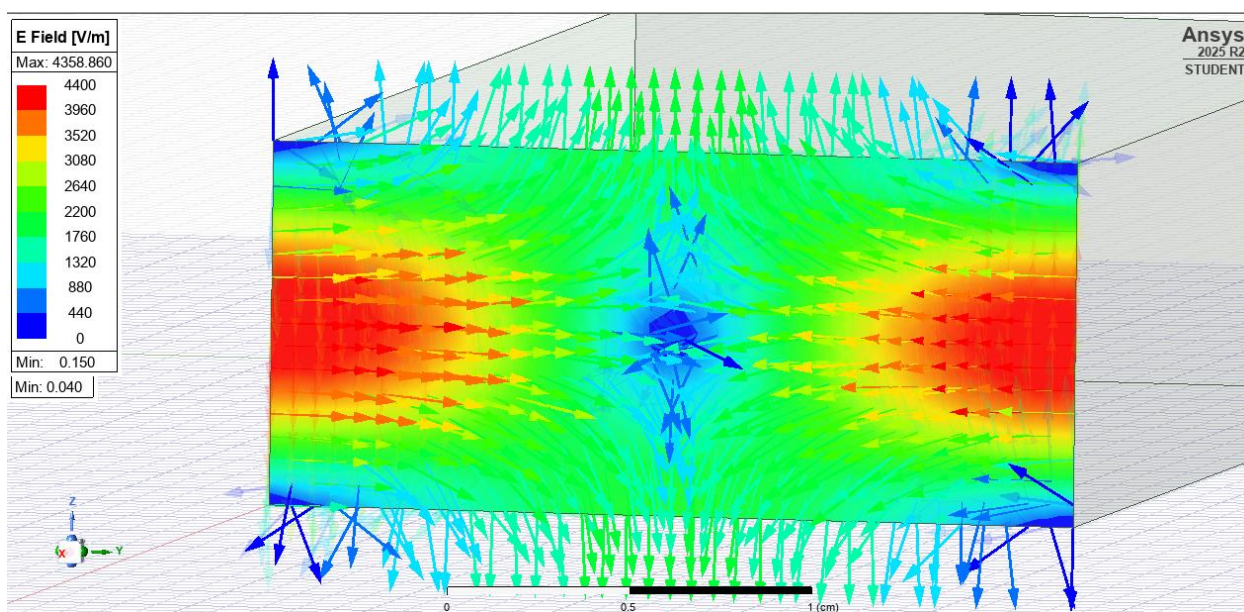


Figure 22 - Port Face E-field distribution Mode 4

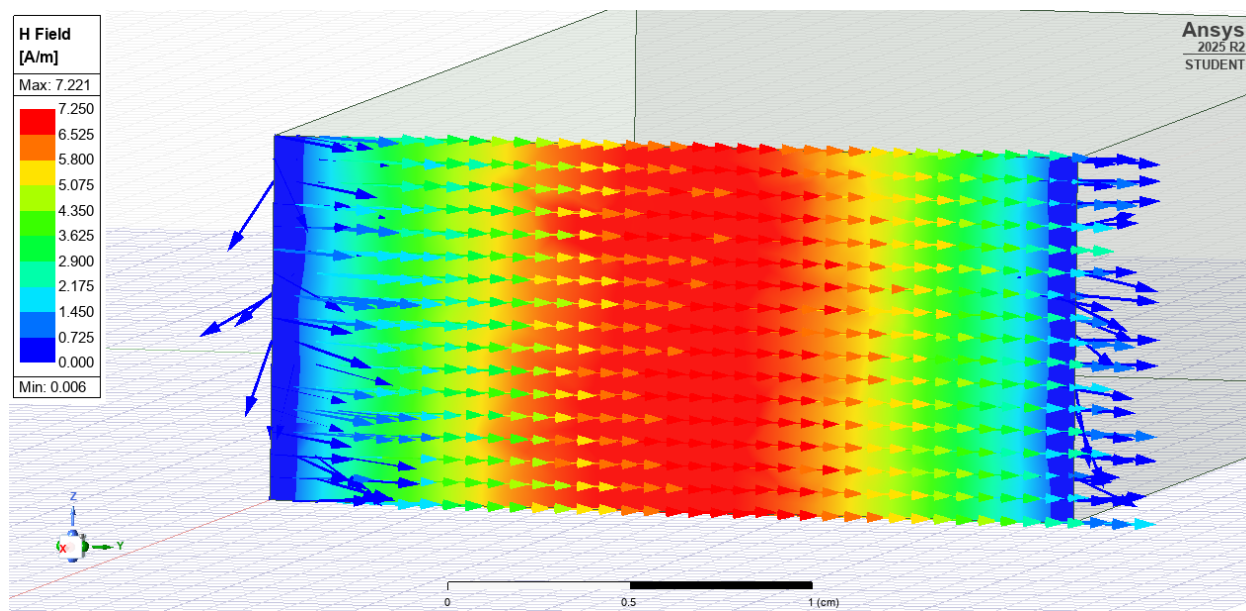


Figure 23 - Port Face H-field distribution Mode 1

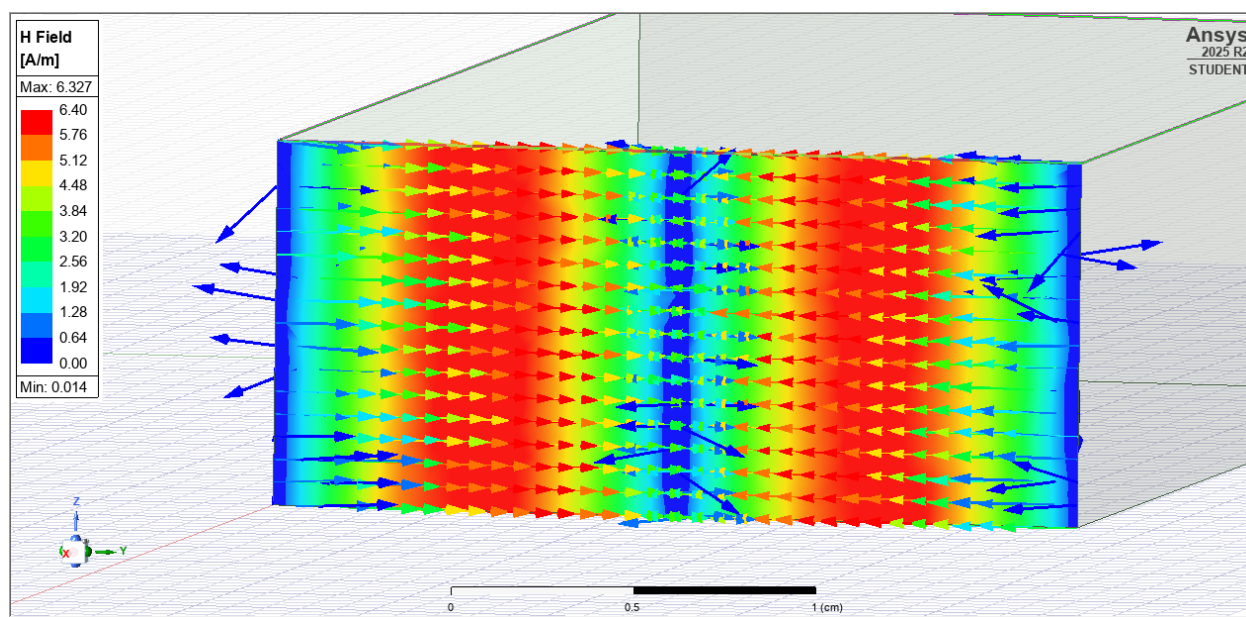


Figure 24 - Port Face H-field distribution Mode 2

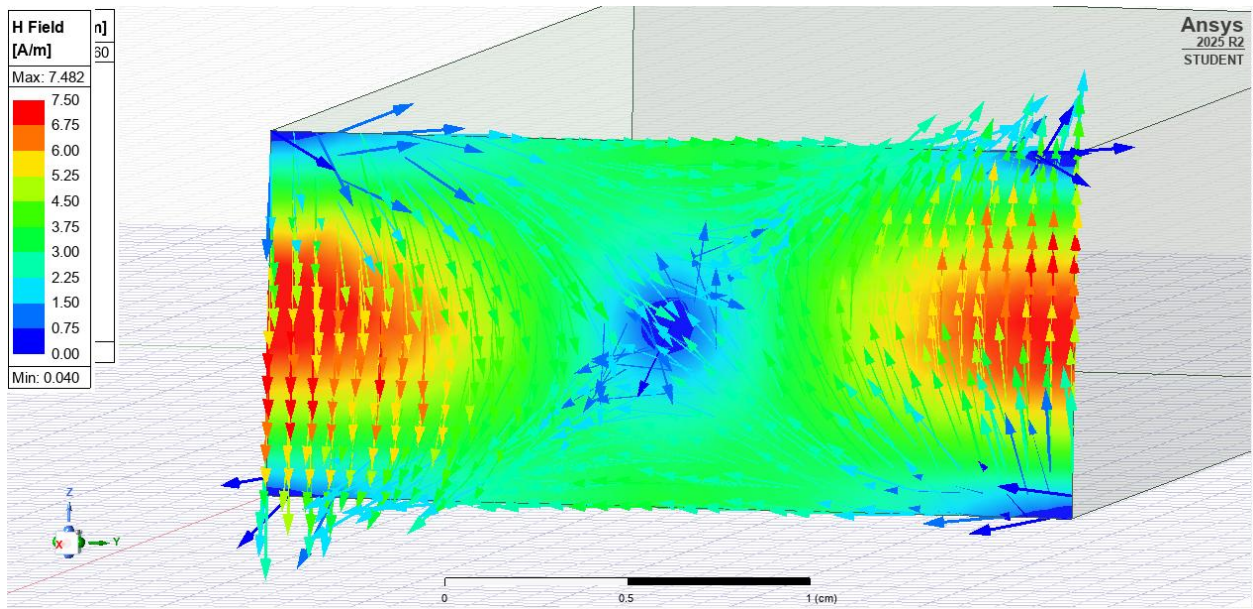


Figure 25 - Port Face H-field distribution Mode 4

C: HFSS Settings

The following figures provide a detailed look at the settings used in HFSS for simulations throughout the project.

Driven Solution Setup

General | Mesh/Solution Options | Adaptive Options | Advanced | Expression Cache | Derivatives | Defaults

Setup Name:

☒ Enabled ☐ Solve Ports Only

Adaptive Solutions

Solution Frequency: ☒ Single ☐ Multi-Frequencies ☐ Broadband

Frequency:

Maximum Number of Passes:

☒ Maximum Delta S:

☐ Use Matrix Convergence:

Use Defaults

HPC and Analysis Options...

OK Cancel

Figure 26 – Setup Type and Description

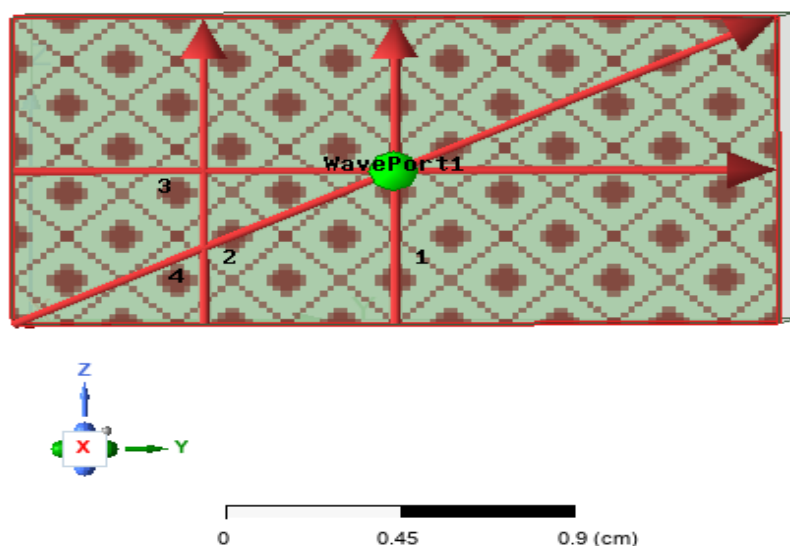


Figure 27 – Excitation characteristics for the four modes of operation

Edit Frequency Sweep ✕

General | Defaults

Sweep Name: ☒ Enabled

Sweep Type: ▼

Frequency Sweeps [33 points defined]

	Distribution	Start	End		
1	Linear Step	4GHz	20GHz	Step size	0.5GHz

3D Fields Save Options

☐ Save Fields (All Frequencies)

☐ Save radiated fields only

Time Domain Calculation...

S Matrix Only Solve

☒ Auto

☐ Manual - Allow for frequencies above

▼

Figure 28 – Settings for Frequency Sweep (Part 1)

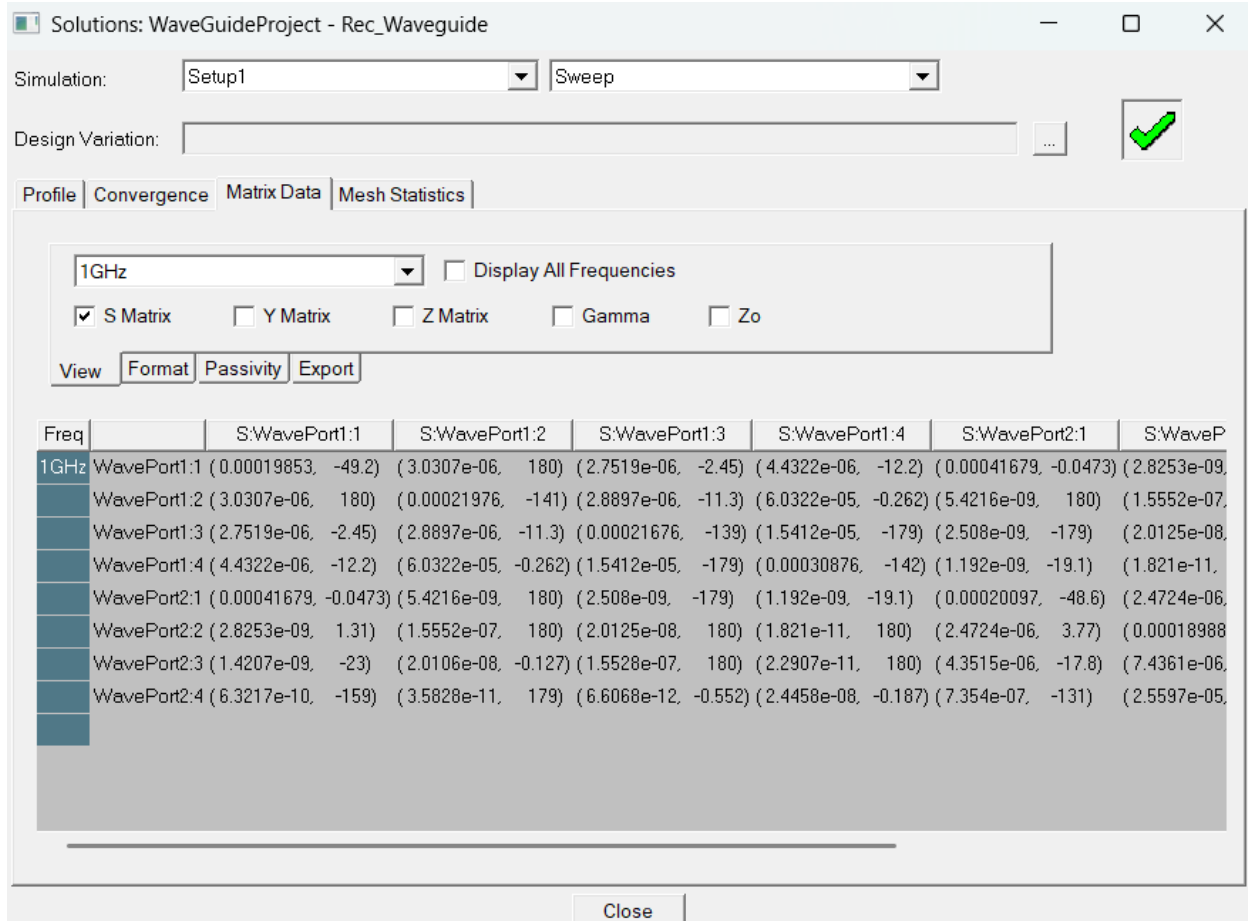


Figure 29 – Solution data S-parameter

[Rec_Waveguide] Edit post process sources

Spectral Fields | Source Context

	Source	Type	Magnitude	Unit	Phase	Unit
1	WavePort1.1	Port	0 W		0 deg	
2	WavePort1.2	Port	1 W		0 deg	
3	WavePort1.3	Port	0 W		0 deg	
4	WavePort1.4	Port	0 W		0 deg	
5	WavePort2.1	Port	0 W		0 deg	
6	WavePort2.2	Port	0 W		0 deg	
7	WavePort2.3	Port	0 W		0 deg	
8	WavePort2.4	Port	0 W		0 deg	

☐ Include Port Post Processing Effects
 ☐ Weighted Element Pattern Post Processing Mode

System power for gain calculations:

☐ Specify System Power: W

☒ Use Maximum Available Power

Figure 30 – Selecting the mode of operation for field distribution simulations (Mode 2 selected)